CHAPTER -06
APPLICATION OF DERIVATIVES

MCQ /FB questions:

1. The rate of change of the area of a circle per second with respect to its radius r when r = 5cm is.... .... cm^2/s.
   a)10π b)12π c)8π d)11π
2. The rate of change of the area of a circle per second with respect to its radius r when r = 4 cm is.... ....$\pi cm^2/s$.
   a)10 b)12 c)8 d)11
3. The total revenue Rupees received from the sale of x units of a product is given by R (x)=$13x^2+26x-15$, then the marginal revenue when $x=1$ is ..... Rupees
   a)26 b)13 c)52 d)104
4. The total revenue in rupees received from the sale of x units of a product is R (x) = $3x^2+36x+5$, then the marginal revenue when $x=15$ is ..... Rupees
   a)116 b)96 c)90 d)126
5. The radius of circle is increasing at the rate of 0.7 cm/sec, then the rate of increase of its circumference is..... π cm/sec
   a)2 b)1.4 c)0.7 d)4.9
6. The radius of an air bubble is increasing at the rate of $\frac{1}{2}$cm/s, then the volume of the bubble increasing when the radius is 1 cm is .....cc/sec
   a)2π b) 2 c) 2π d) 8π
7. The function $f(x)=\cos x$ is increasing in the interval.....
   a)$\left(0,\frac{\pi}{2}\right)$ b)$(0,\pi)$ c) $\left(\frac{\pi}{2},\pi\right)$ d)$(\pi,2\pi)$
8. The function $f(x)=3x+17$ is strictly increasing on .....
   a)$(-\infty,\infty)$ b)$(0,\infty)$ c)$(-\infty,0)$ d)$(0,3)$.
9. The interval in which $y=x^2e^{-x}$ is increasing is.....
   a)$(-\infty,0)$ b)$(-\infty,0)\cup(2,\infty)$ c)$(2,\infty)$ d)$(0,2)$
10. The interval in which $y=x^2e^{-x}$ is decreasing is.....
    a)$(-\infty,0)$ b)$(-\infty,0)\cup(2,\infty)$ c)$(2,\infty)$ d)$(0,2)$
11. The minimum value of $|x|$ in R is.....
    a)0 b)1 c)2 d)*not exists*.
12. The maximum value of the function $f(x)=|x|$ *is.... in R.*
    a)0 b)1 c)2 d)*not exists*.
13. The maximum values of the function given by $f(x)=x,\ x\in[0,1]$ is .....
    a)0 b)1 c)2 d)*not exists*.
14. The minimum values of the function given by $f(x)=x,\ x\in[0,1]$ is .....
    a)0 b)1 c)2 d)*not exists*.
15. In second derivative test if $f'(c)=0$ *and* $f''(c)>0$ for some 'c' on an interval 'I', then the function f attains.....
    a) absolute maximum value b)absolute minimum value
    c) local maximum value. d)local minimum value.
16. A stone is dropped into a quiet lake and waves move in circles at a speed of 4cm per second. At the instant, when the radius of the circular wave is 10 cm, ... cm^2/sec fast is the enclosed area increasing.
    a)20π b)40π c)80π d)60π
17. Point of minimum value of the function given by $f(x)=|x|$ *is....*
    a)1 b)2 c)0 d)*not exists*.
18. The number of points of local maxima and local minima of the function f given by $f(x)=x^3-3x+3$ *is....*
    a)0 b)1 c)2 d) 3.
19. The absolute maximum value of the function f given by $f(x)=x^3$,$x\in[-2,2]$ *is* =...
    a)-2 b)2 c)0 d)8.

20. The total cost C(x) in Rupees, associated with the production of x units of an item is given by C(x) = 0.005 x^3 – 0.02 x^2+ 30x + 5000.Then the marginal cost when 3 units are produced, where by marginal cost we mean the instantaneous rate of change of total cost at any level of output....

a)30.135 b)30.255 c)30.015 d)30.

21. The function f given by $f(x) = x^2 - x + 1$ is.

a) neither strictly increasing nor decreasing on (– 1, 1) b) decreasing on $\left(\frac{1}{2},\infty\right)$

c) increasing on (–∞,∞) d)increasing on $\left(-\infty,\frac{1}{2}\right)$

22. Which of the following functions is decreasing on $\left(0,\frac{\pi}{2}\right)$

a)cos x b) sinx c) cos $3x$ d) tan x

23. The values of a=..... for the function f given by $f(x) = x^2 + ax + 1$ is increasing on [1, 2]

a) a<-2 b)$a < 2$ c)$a > -2$ d)$a < 2$.

24. The function f given by $f(x) = log|sinx|$is increasing on

a)$\left(0,\frac{\pi}{2}\right)$ b)$(0,\pi)$ c) $\left(\frac{\pi}{2},\pi\right)$ d)$\left(\frac{3\pi}{2},2\pi\right)$

25. A cylindrical tank of radius 10 m is being filled with wheat at the rate of 314cubic metre per hour. Then the depth of the wheat is increasing at the rate of

a)1 m/h b) 0.1 m/h c) 1.1 m/h d) 0.5 m/h

26. The maximum and minimum values of the function $|sin4x + 3|$ are

a)1,2 b)4,2 c)2,4 d)-1,1

27. Which of the following functions is decreasing on $\left(0,\frac{\pi}{2}\right)$

a) $sin2x$ b) $tanx$ c)$cosx$ d) $cos\ 3x$

28. On which of the following intervals is the function f given by $f(x) = x^{100} + \sin x - 1$ decreasing ?

a)$\left(0,\frac{\pi}{2}\right)$ b)(0,1) c) $\left(\frac{\pi}{2},\pi\right)$ d) none of thess.

29. If a function f is such that $f^{/}$ (c) = 0 and f $f^{//}$(c) < 0 for some 'c' on an interval 'I', then at c the function f attains its

A) absolute maximum value B) absolute minimum value

C) local maximum value D) local minimum value

30. The edge of a cube is increasing at the rate of $5cm/\sec$. How fast is the volume of the cube increasing when the edge is $12cm$ long

(a) $432cm^3/\sec$ (b) $2160cm^3/\sec$ (c) $180cm^3/\sec$ (d)None of these

31. The radius of an air bubble is increasing at the rate of 1cm/s, then the volume of the bubble increasing when the radius is 1cm is ..... π cc/sec

32. Minimum value of f(x) = $|x + 2| - 1$ is----------

33. Local maxima of f(x) = sin x + cos x, 0 <x<π is ----------

34. If $x$is a real, the minimum value of $x^2 - 8x + 17$ is ----------

35. The maximum value of $sinx.cosx$ is ----------

36. If $x + y = 10$, then the maximum value ofxy-------

37. The absolute maximum value of the function f given by $f(x) = x^3, x \in [-2, 2]$ *is*---------

38. **Assertion (A) :** The functionf(x) = x^2 is decreasing in the interval (0, ∞)

Reason (R) : Any function y = f(x) is decreasing, if $\frac{dy}{dx} < 0$

A) A is false but R is true B) A is false and R is false

C)A is true but R is false D) A is false and R is false.

39. **Statement-I :** Intervals in which the function $f(x) = x^2 - 4x + 6$ is strictly increasing is (2, ∞).

Statement-2: Let f be continuous on [a,b] and differentiable on the open interval (a,b). Then f is strictly increasing function in [a,b] if f '(x) > 0 for each x ∈ (a,b).

A) Statement 1 is true and Statement 2 is false.

B) Statement 1 is true and Statement 2 is true, Statement 2 is correct explanation for Statement 1

C) Statement 1 is true and Statement 2 is true, Statement 2 is not a correct explanation for Statement 1

D) Statement 1 is false and Statement 2 is false.

40. **Assertion (A):** A particle moving in a straight line covers a distance of x cm in t seconds , where $x = t^3 + 3t^2 - 6t + 18$. The velocity of particle at the end of 3 seconds is 9cm/s

Reason (R): Velocity of the particle at the end of 3 seconds is $\frac{dx}{dt}$ $at\ t = 3$

A) A is false but R is true B) A is true and R is false

C) A is true but R is true D) A is false and R is false

41. **Statement-1:** The function f (x) = $x^3 - 12x$ is strictly increasing in $(-\infty, -2) \cup (2, \infty)$.

Statement-2: For strictly increasing function f in an interval I , f ′ (x) > 0 for all $x \in I$.

A) Statement 1 is true and Statement 2 is false.

B) Statement 1 is false and Statement 2 is true,

C) Statement 1 is true and Statement 2 is true,

D) Statement 1 is false and Statement 2 is false.

42. **Assertion (A):** The maximum value of the function f(x) = x^5 , x ∈ [–1, 1], is attained at its critical point, x = 0.

Reason (R): The local maximum or local minimum values of a function can only occur at turning points .

A) A is false but R is true B) A is false and R is false

C)A is true but R is false D) A is false and R is false.

43. **Statement-1:** The rate of change of area of a circle with respect to its radius r when r = 6 cm is 12πcm² /cm.

Statement-2 : Rate of change of area of a circle with respect to its radius r is $\frac{dA}{dr}$, where A is the area of the circle.

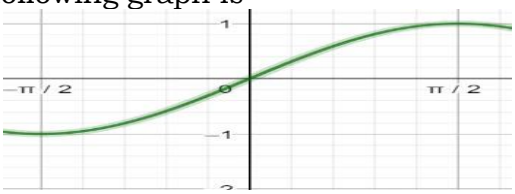
A) Statement 1 is true and Statement 2 is false.

B) Statement 1 is true and Statement 2 is true, Statement 2 is correct explanation for Statement 1

C) Statement 1 is true and Statement 2 is true,Statement 2 is not a correct explanation forStatement 1

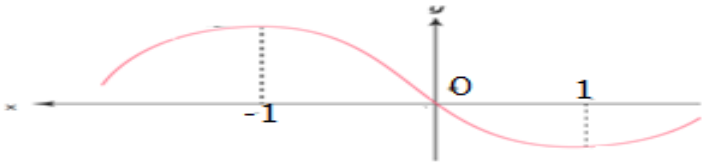
D) Statement 1 is false and Statement 2 is false.

44. The point of inflection for the following graph is



(A)$-\frac{\pi}{2}$ B)$\frac{\pi}{2}$ C) 0 D) point of inflection does not exist

45. The point of local maxima, local minima and inflection for the following graph respectively is



A)1,0, −1 B) 0 − 1, 1 C) -1, 0 ,1 D) -1,1,0

46. ***Assertion (A)*** **:**The maximum value of the function $f(x) = x^3$, $x \in [-1, 1]$, is attained at its end point, $x = 1$.

Reason (R)**:** The function $f(x) = x^3$ is strictly increasing function in [-1, 1].

A)Both A and R are true and R is the correct explanation for A

B) Both A and R are true but R is not the correct explanation for A.

C) A is false but R is true. D) Both A and R are false.

47. The absolute maximum value of y = $x^3 - 3x + 2$ in 0 ≤ x ≤ 2 is

A) 0 B) 2 C) 4 D) 6

48. The function f(x) = x + cos x is

A) Always increasing B) Always decreasing

C)Increasing for a certain range of x D)None

49. Let the f: R → R be defined by f(x) = 2x + cos x, then

1. f has a maximum value at x = 0
2. f is an increasing function
3. f is a decreasing function

Which of the above statements are correct?

(A) 1 only (B) 2 only (C) 3 only (D) All 1, 2 and 3.

50. Consider the function 1.f(x) $= e^{-x}$ 2. f(x) $= x^2 - sinx$ 3. f(x) $= \sqrt{x^3+1}$

Which of the above functions is/ are increasing in [0,1]

(A) 2 only (B) 2 and 3 only (C) 3 only (D) 1 and 3 only.

51. The point(s) on the curve y = x^2, at which y-coordinate is changing four times as fast as x- coordinate is/are

(A) (4, 16) (B) (2, 4) (C) (-2, 4) (D) (2, 4), (-2, 4).

52. The function f(x) = $x^5 - 5x^4 + 5x^3 - 1$ has

(A) 1 critical point (B) 2 critical points (C) 3 critical points (D) 4 critical points

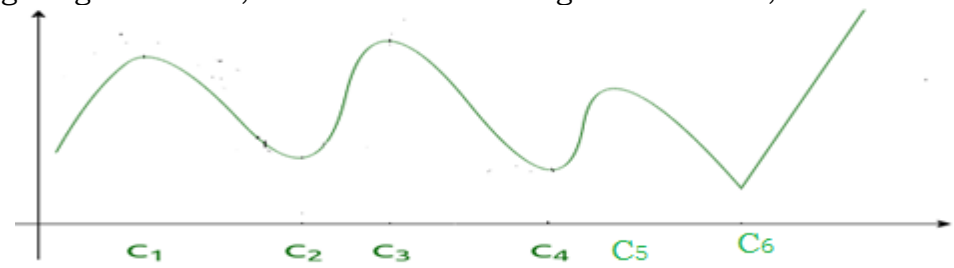
53. The volume of a sphere is increasing at the rate of $\pi cm^3/sec$. The rate at which the radius is increasing is_____, when the radius is 3cm.

(A) $\frac{1}{36} cm/sec$ (B) $36cm/sec$ (C) 9 cm/sec (D) $27cm/sec$

54. The function f(x) = x^x decreasing in .

(A) (0, e) (B) (0, ∞) (C) (0, $\frac{1}{e}$) (D) (0, 1).

55. For the figure given below, consider the following statements 1,2 and 3



Statement 1: f has local maximum values at x = C_1, C_3, C_5.

Statement 2: f has local minimum values at x = C_2, C_4, C_6.

Statement 3: f has neither local maximum nor local minimum value at x = C_6.

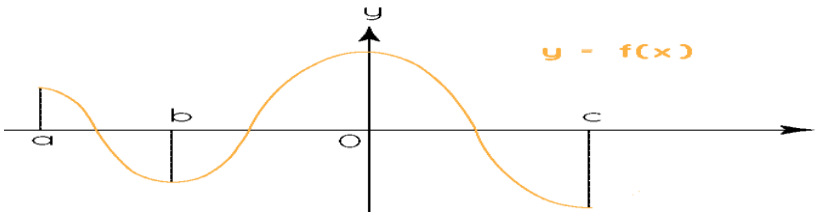
A) Statement 1 is true and Statement 2 and 3 are false.

B) Statement 1 and 2 are true but Statement 3 is false.

C) Statement 1 and 3 are true but Statement 2 is false.

D) Statement 2 and 3 are true but Statement 1 is false.

56. For the figure given below, consider the following statements 1 and 2



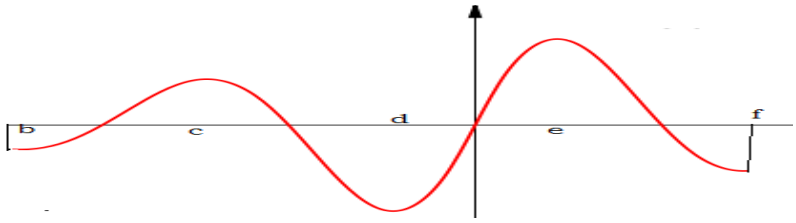
Statement 1: The maximum value of the function $y = f(x)$, $x \in$ [a, c], is f(0)

Statement 2: The minimum value of the function $y = f(x)$, $x \in$ [a, c], is f(b)

A) Statement 1 is true and 2 is false. B) Statement 1 is true and 2 is true

C) Statement 1 and 2 are false. D) Statement 1 is false and 2 is true.

57. For the figure given below, consider the following statements 1 and 2



Statement 1: The local maximum value of the function $y = f(x)$, $x \in$ [b, f], is f(c)
Statement 2: The absolute maximum value of the function $y = f(x)$, $x \in$ [b, f], is f(e)
A) Statement 1 is true and 2 is false. B) Statement 1 and 2 are true
C) Statement 1 and 2 are false. D) Statement 1 is false and 2 is true.

58. The maximum and minimum values of the function $|sin4x + 3|$ are
A)1,2 B)4,2 C)2,4 D)-1,1

59. If x is real, the minimum value of $x^2 - 8x + 17$ is
A) -1 B) 0 C) 1 D) 2

60. The function $f(x) = tanx - x$
A) always increases B) always decreases
C) never increases D) sometimes increases and sometimes decreases.

61. The maximum value of $3sinx + 4\,cosx$ is
A) 25 B) 5 C)$\sqrt{5}$ D) $\sqrt{7}$.

62. **Statement 1:** The function $y = e^{-x}$ is ever increasing in the set of real numbers R.
Statement 2 : The function $y = log_e x$ is ever decreasing in $(0, \infty)$.
(A) Statement 1 is false and Statement 2 is true
(B) Statement 1 is true and Statement 2 is false
(C) Both Statements 1 and 2 are true (D) Both Statements 1 and 2 are false.

63. Consider the function f *given by* $f(x) = -x^2$
Statement 1: The function f is strictly increasing in $(-\infty, 0)$.
Statement 2 : The function f is neither increasing nor decreasing in $(-\infty, \infty)$.
(A) Statement 1 is false and Statement 2 is true
(B) Statement 1 is true and Statement 2 is false
(C) Both Statements 1 and 2 are true (D) Both Statements 1 and 2 are false.

64. Consider the function f *given by* $f(x) = (x-1)^3$
Statement 1: $x = 1$ is a point of inflection of f.
Statement 2: An interior critical point which is neither a point of local maxima nor a point of local minima is a point of inflection.
(A) Statement 1 is false and Statement 2 is true
(B) Statement 1 is true and Statement 2 is false
(C) Statement 1 is true, statement 2 is true and statement 2 is a correct explanation for statement 1
(D) Statement 1 is true , statement 2 is true and statement 2 is not a correct explanation for statement 1.

65. Consider the function f *given by* $f(x) = |x| + 3$
Statement 1: *The critical point of* f *is* 0. .
Statement 2: Local minimum value of f *at* $x = 0$ *is* 3.
(A) Statement 1 is false and Statement 2 is true
(B) Statement 1 is true and Statement 2 is false
(C) Both Statements 1 and 2 are true
(D) Both Statements 1 and 2 are false